

Vinal Auto Traders – Project Report

1. Introduction

This project is about developing an online Vehicle Traders Website named Vinal Auto Traders. The main goal of this system is to provide a complete online platform for vehicle trading in Sri Lanka. Customers can view vehicles, compare them, order parts, book test drives, submit reviews, and contact the company. The website also includes an Admin Panel, where administrators can manage all the activities of the website such as vehicles, parts, users, messages, orders, and bookings.

The project was developed as a university group assignment for the Web Application Development module. The system was built using PHP as the backend, MySQL as the database, and HTML, CSS, and JavaScript for the frontend. GitHub was used for project version control and team collaboration.

The website is designed in a user-friendly way so that even a new customer can easily browse vehicles, compare options, and book services. At the same time, the admin panel is built with strong security features such as login authentication, OTP verification, password reset with PHPMailer and Gmail SMTP, and role-based user levels.

Overall, the website provides a complete online solution for vehicle trading, including chatbot support, financial calculator, Google Maps integration, and live booking notifications.

2. Objectives

The main objectives of the Vinal Auto Traders Website are:

- To develop a complete business website for a Vehicle Traders company.
- To allow customers to view available vehicles with details and images.
- To provide a compare feature so that customers can compare two or more vehicles.
- To implement a chatbot that can answer customer questions and guide them in vehicle selection.
- To allow customers to book test drives using online forms.
- To enable customers to place vehicle or parts orders directly through the system.
- To provide a wishlist feature for saving vehicles for later viewing.
- To implement reviews and feedback submission for customers.
- To allow secure admin login with options for OTP, Gmail SMTP password reset, and strong password rules.
- To provide an admin panel where admins can manage vehicles, parts, orders, bookings, users, and customer messages.
- To create a financial calculator that helps customers calculate easy monthly plans and loan amounts.
- To integrate Google Maps API to show dealership locations and directions.

- To create a subscribe system where customers can subscribe with their email to receive updates.
- To use PHPMailer to send confirmation emails from forms using SMTP.
- To make use of GitHub for collaboration, contribution tracking, and version control.
- Vehicle PDF Reports: The admin panel allows administrators to generate and download PDF reports of all vehicles in the inventory.
- Vehicle Comparison: Customers can compare two or more vehicles side by side, viewing details like engine type, price, fuel type, and performance.
- Financial Guidance Chatbot: The website includes a chatbot that can guide customers on financial options, helping them calculate monthly payments and loan plans.

3. Website Features (User Side)

The website contains several main pages and features that make it easy for customers to interact with the company.

3.1 Home Page

- A welcoming page with the company slogan:
“Find your dream ride with trusted dealers. From sleek city cars to rugged off-roaders, we bring you the best deals in Sri Lanka.”
- Highlights featured vehicles.
- Shows sections like Wide Selection, Trusted Dealers, and Smart Financing.
- Includes a chatbot button for customer support.

3.2 Vehicles Page

- Displays a full list of vehicles with details like make, model, year, price, and images.
- Includes filters for vehicle type (hatchback, sedan, SUV, pickup).
- Customers can sort vehicles by price, popularity, or newest.
- Vehicles have options like View Details, Add to Wishlist, and Compare.

3.3 Compare Vehicles

- Customers can select two or more vehicles to compare.
- The system shows a side-by-side table with details such as engine, price, fuel type, and performance.
- This feature helps customers make better buying decisions.

3.4 Parts Page

- Displays available vehicle parts and accessories.
- Customers can order parts directly through the website.
- Includes messages to the admin if parts are not available.
- Admin can also reply to parts requests.

3.5 Book Test Drive

- Customers can book test drives online.
- A form collects details like name, contact, vehicle selected, and date.
- The booking is stored in the database and an email notification is sent using PHPMailer.
- Customers receive confirmation, and admins can update booking status.

3.6 Orders

- Customers can place vehicle or parts orders.
- Orders are stored in the database and linked to customer accounts.

3.7 Wishlist

- Customers can add vehicles to a wishlist.
- They can later view or remove vehicles from the wishlist.

3.8 Reviews

- Customers can leave reviews and ratings.
- Reviews are displayed on the vehicle or parts page.
- Admin has the power to approve, delete, or reply to reviews.

3.9 Contact Us Page

- A contact form that allows customers to send messages.
- Uses PHPMailer with Gmail SMTP to send the message to the company email.
- Confirmation email is sent back to the customer.

3.10 Chatbot

- Interactive chatbot built using simple PHP logic and JavaScript.
- Helps answer common questions such as:

ex:

"What vehicles are available?" "How can I book a test drive?" "What financing options do you provide?"

- **Can also guide customers to the Compare Vehicles page.**

3.11 Financial Calculator

- Allows customers to calculate monthly payment plans.
- Customers enter vehicle price, loan years, and interest rate.
- The system calculates monthly installment.

3.12 Subscription System

- Customers can subscribe with email.
- The system stores the email in the database.

- Admin can send updates to subscribers about new offers.

3.13 Google Maps Integration

- Shows dealership locations on a map.
- Helps customers easily find the nearest branch.

4. Admin Panel Features

The admin panel is a secure part of the system with several management tools.

4.1 Login and Authentication

- Secure login system using PHP sessions.
- Passwords are hashed and stored securely.
- Admin can reset passwords using OTP via email (PHPMailer + Gmail SMTP).
- Strong password rules (minimum length, special characters).

4.2 Dashboard

- Shows summary of website activities.
- Displays number of vehicles, bookings, users, and pending messages.

4.3 Manage Vehicles

- Admin can add, edit, and delete vehicles.
- Vehicle details include name, model, year, price, and description.
- Vehicle images can also be uploaded.
- Vehicle report PDF generation

4.4 Manage Parts

- Admin can add, edit, and delete parts.
- Can reply to customer part requests.
- Can view all part orders and messages.

4.5 Manage Users

- Admin can add different levels of users (Admin, Staff, Customer).
- Staff users can manage bookings and orders but cannot delete vehicles.
- Secure system with role-based access.

4.6 Manage Orders and Bookings

- Admin can view all orders and bookings.
- Can update booking status (approved, cancelled).
- Customers are notified by email about status changes.

4.7 Messages

- Admin can view and reply to customer messages.
- Replies are sent using PHPMailer.

4.8 Reviews Management

- Admin can view all reviews.
- Admin can approve or delete inappropriate reviews.

4.9 Security

- OTP system for password recovery.
- Password reset link sent via email.
- Strong password enforcement.

5. Database Details

The system uses a MySQL database named vinal_auto.

Tables

1. users – Stores user accounts with roles (admin, staff, customer).
2. vehicles – Stores all vehicle details.
3. parts – Stores parts details.
4. orders – Stores customer orders.
5. bookings – Stores test drive bookings.
6. reviews – Stores customer reviews.
7. messages – Stores contact messages.
8. compare – vehicles compare customers.
9. subscriptions – Stores subscriber emails.

Each table supports DML operations (INSERT, UPDATE, DELETE, SELECT).

6. Tools and Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Email: PHPMailer with Gmail SMTP
- Version Control: Git and GitHub
- Server: cPanel Hosting / Localhost (XAMPP)
- APIs: Google Maps API for location
- Security: PHP Sessions, OTP, Password Hashing
- grhaph and Chart.js, Canvas, JavaScript graphs
- Vehicle Report PDF **FPDF** library use
- use librarys

7. Security Features

- Strong password requirements.
- OTP verification during password reset.
- PHPMailer with Gmail SMTP for secure emails.
- User roles for access control.
- SQL Injection protection with prepared statements.
- Session handling for secure login/logout.

8. Special Features

- Chatbot for customer support.
- Vehicle Compare to help customers choose.
- Financial Calculator for easy loan calculations.
- Google Maps API integration for locations.
- Email Subscription System for updates.
- Wishlist and Reviews for better user experience.

9. Deliverables

- Deliverable 1 - Developed website (PHP + MySQL).
- Deliverable 2 - Project report with GUI screenshots and database details.
- Deliverable 3 - Group presentation with system demonstration.

10. Conclusion

The Vinal Auto Traders Website is a complete online solution for vehicle sales and management. It allows customers to view, compare, and book vehicles while also interacting with the company through chatbots, forms, and reviews.

The admin panel ensures smooth management of vehicles, parts, users, and bookings. Security is handled with OTP, PHPMailer, and strong password rules. Additional features such as financial calculator, Google Maps, and subscription system make the platform modern and useful.

The project was successfully developed using PHP, MySQL, and GitHub collaboration. It meets the objectives of the assignment and provides a strong foundation for real-world use by a Vehicle Traders company.

GROUP AZ

Member	ST -ID
THISAL CHATHNUKA	36224
RANITHU SENSITH	36453
ANUPAMA JAYAWICKRAMA	36189
ROSHAN SAMPATH	37203
VIDATH ALAHAKOON	36193
ACHINTHA INDRAJITH	36904
THATHSARA PRAMUDITHA	36481
SANDEEPA HANSTHA	36524
JEEWA LIYANAARACHI	36524
W.N.T.Alwis	36224

GIT HUB LINK

https://github.com/Thisal00/vinal_CAR_WEB.git

WEB HOSTING LINK

<https://thisal.slhosted.lk>

ADMIN LOGIN

user name - admin

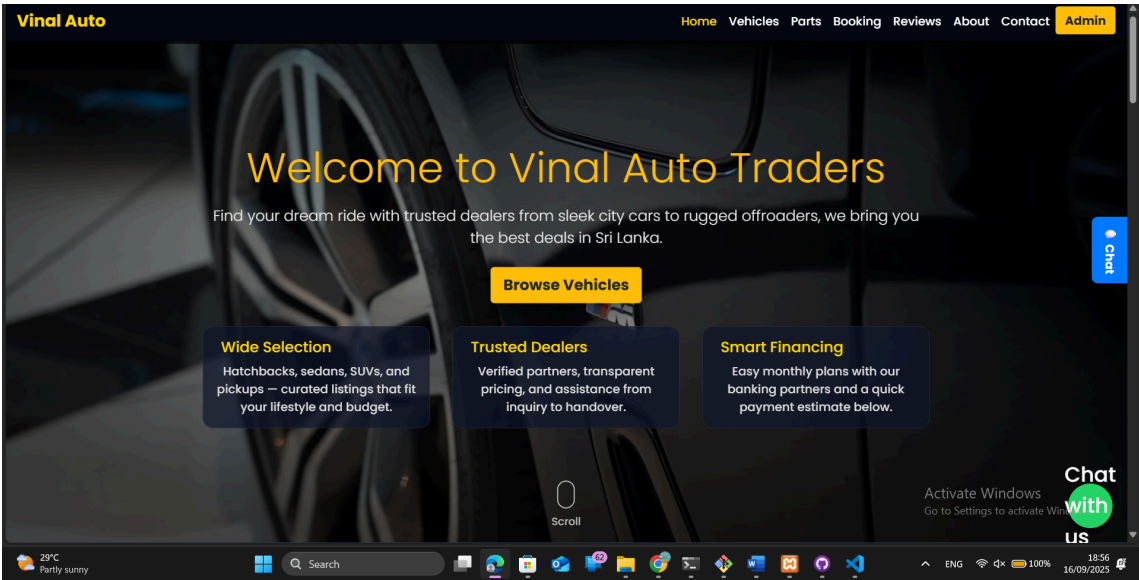
password - 20040224

FULL FILE LINK

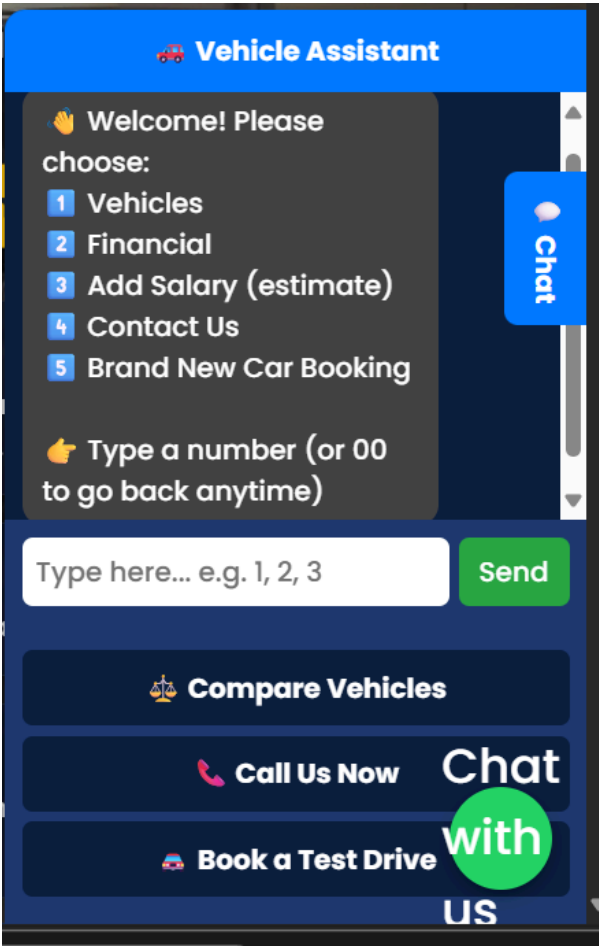
https://nsbm365-my.sharepoint.com/:u:/g/personal/wtchathnuka_students_nsbm_ac_lk/EQS5w9AYGV5FhoGqAwkTl14Bxs_RLVPXmSUdO9bDUd7zXQ?e=AKgXGg

UI Screenshots

01



02



Vinal Auto

Home

Vehicles

Parts

Booking

Reviews

About

Contact

Admin

Vehicle Comparison Tool

Select First Vehicle:

TATA

Select Second Vehicle:

COROLLA

Compare

Comparison Table

Feature	COROLLA	TATA
Fuel Efficiency (km/l)	50	10
Price (New)	Rs. 1,000,000	Rs. 1,000,000
Resale Value	Rs. 1,000,000	Rs. 10,000

Feature Comparison Chart

Activate Windows
Go to Settings to activate Windows.

TOYOTA

H

ISUZU

LAND-ROVER

MG

TOYOTA

Finance Calculator

Amount (LKR)

1000000

Down Payment (LKR)

10000

Interest Rate (%)

12

Loan Term (Months)

10

Calculate

Reset

Monthly Payment: 104,526 LKR

Estimate only. Actual figures depend on bank evaluation.

BOC

Seylan

HNB HNB

LB Finance

Siyapatha

NSB

BOC

Activate Windows
Go to Settings to activate Windows.

Vinal Auto

Vehicles

Home

Vehicles

Parts

Booking

Reviews

About

Contact

Admin

Make or Model

Min Price

Max Price

Min Year

Max Year

Fuel

Transmission

Filter

2025 BYD atto 3

LKR 22,500,000.00

Hybrid • Automatic • 10,000 km

Brand: BYD Model: ATTO 3 Trim / Edition: 2024 Year of Manufacture: 2024 Condition: New

Transmission: Automatic Body type: SUV / 4x4...

View Details

2018 Mercedes Benz C200 AMG Premium Plus

LKR 31,500,000.00

Petrol • Automatic • 20,000 km

Mercedes Benz C200 AMG Line Premium Plus 2018 32,000Km Done Agent Maintained Pearl white Exterior Tan Leather Interior Both Side Electric &...

View Details

2024 Toyota Raize Z Grade

LKR 15,850,000.00

Hybrid • Manual • 10,000 km

Toyota Raize Z Grade 2024 Highest Spec Brand New Unregistered Auction Grade: S (brand New) Engine: 1200cc Petrol Automatic Transmission...

View Details

Activate Windows
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Vinal Auto
Available Vehicle Parts
Home Vehicles Parts Booking Reviews About Contact Admin

Search part name...
Search



AC Cooler
Rs. 13,000.00

Condition: Reconditioned Part or Accessory Type: Air Conditioning Heating Brand: Other brand M...

View Email WhatsApp



Daihatsu Cast Front Door (Doors)
Rs. 10,000.00

Condition: Used Part or Accessory Type: Doors Brand: Daihatsu Model: Other Model

View Email WhatsApp



BMW X5 Alternator
Rs. 150,000.00

Condition: New Part or Accessory Type: Engines & Engine Parts Brand: BMW Model: X5

View Email WhatsApp



Honda Vezel RS Seat Set
Rs. 325,000.00

Condition: Reconditioned Part or Accessory Type: Interior Parts & Furnishings Brand: Honda ...

View Email WhatsApp









Vinal Auto
Book a Test Drive

Your Name:
THISAL CHATHNUKA
Email Address:
thisalchathnuka@gmail.com
Phone Number:
0768291088
Vehicle:
BMW X5 M E40 MSport
Date:
25/09/2025
Time:
20:42
Book Test Drive

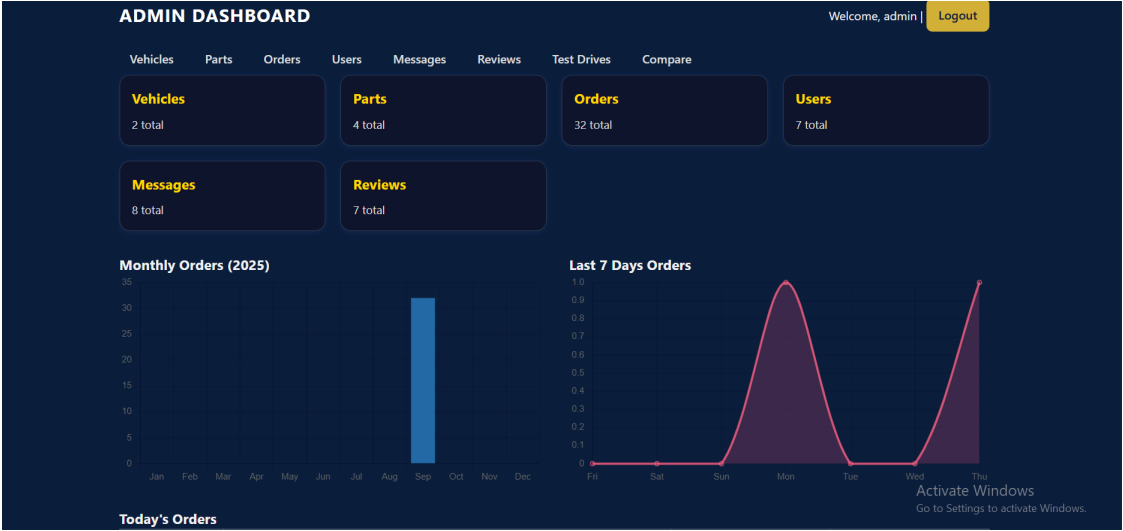
Gmail
Search mail
1 of 5
to me
Someone just submitted your form on https://thisal.sihosted.lk/
Here's what they had to say:
message:
Hello THISAL CHATHNUKA,
Your test drive booking request has been received!
Vehicle ID: 87 Date: 2025-09-25 Time: 20:42
We will confirm your booking shortly.
Thanks,
Vinal Auto Team
Submitted at Thu, Sep 18, 2025 9:12 AM (UTC)
Sponsor
Get web development services from an industry expert, h
Activate Windows
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Rs. 10,000.00

Description:
Condition:
Used
Part or Accessory Type:
Doors
Brand:
Daihatsu
Model:
Other Model
Stock: Available (1 left)
Email Admin WhatsApp Admin Buy Now

Name
Phone Number
Email

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PART REQUESTS

Welcome, admin | Logout

VehiclesPartsOrdersUsersMessagesReviewsTest DrivesCompare

#	Part Name	Part ID	Customer Name	Phone	Message	Requested At	Status	Reply	Actions
32	THISAL CHATHNUKA	15	THISAL CHATHNUKA	0768291088	සබ් order එකක් ලැබී ඇත	2025-09-18 10:44:57	unread	No reply	Reply Approve Reject Delete
31	Toyota AC-102EX (Cabin Filter - EX Type)	13	THISAL CHATHNUKA	0768291088	සබ් order එකක් ලැබී ඇත	2025-09-15 12:27:50	approved	No reply	Reply Delete
30	VALVOLINE VR 1 RACING 15W40	14	ranith	+94740611624	සබ් order එකක් ලැබී ඇත	2025-09-10 16:12:02	approved	No reply	Reply Delete
29	Toyota AC-102EX (Cabin Filter - EX Type)	12	ranith	+94740611624	සබ් order එකක් ලැබී ඇත	2025-09-08 14:25:12	unread	No reply	Reply Approve Reject Delete

MANAGE USERS

DashboardLogout

Add User

ShowHideGenerate

staff

Add

Change My Password

ShowHideGenerate

Update

Reset User Password

ShowHideGenerate

Reset

All Users

ID	Username	Role	Actions
15	admin	admin	Delete
13	senali	staff	Delete
6	TC	admin	Delete
5	1234	staff	Delete
4	thisal	admin	Delete
3	ss	staff	Delete

VehiclesPartsOrdersUsersMessagesReviewsTest DrivesCompare

Booking status updated successfully!

Bookings List

Name	Email	Phone	Vehicle	Date	Time	Status	Action
THISAL CHATHNUKA	thisalchathnuka@gmail.com	0768291088	Toyota Voxy SZ	2025-11-09	16:47	Pending	Confirmed Update

Booking Summary

https://mail.google.com/mail/u/...

Gmail

Here's what they had to say:

message:

Hello THISAL CHATHNUKA,

Your test drive booking is confirmed!

Vehicle: Toyota Voxy SZ

Date: 2025-11-09

Time: 16:47

Activate Windows

Go to Settings to activate Windows.

Thanks,

Vinai Auto Team

Add a New Vehicle

[Save Vehicle](#)

List of Vehicles

[Search](#) [Newest](#) [Oldest](#)

ID	Model	Fuel Efficiency	Price	Resale Value	Actions
6	tata	10 km/l	Rs. 1,000,000	Rs. 10,000	Edit Delete
2	COROLLA	50l km/l	Rs. 1,000,000	Rs. 1,000,000	Edit Delete
1	COROLLA	50l km/l	Rs. 1,000,000	Rs. 1,000,000	Edit Delete

Activate Windows
Go to Settings to activate Windows.